ACDC

Understanding coworking members

Exploratory Data Analysis

Data Mining 2026–2027

Group 21

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Data quality changes the interpretation

Objective: understand differences in value, usage and service needs to prepare member segmentation.

15,450

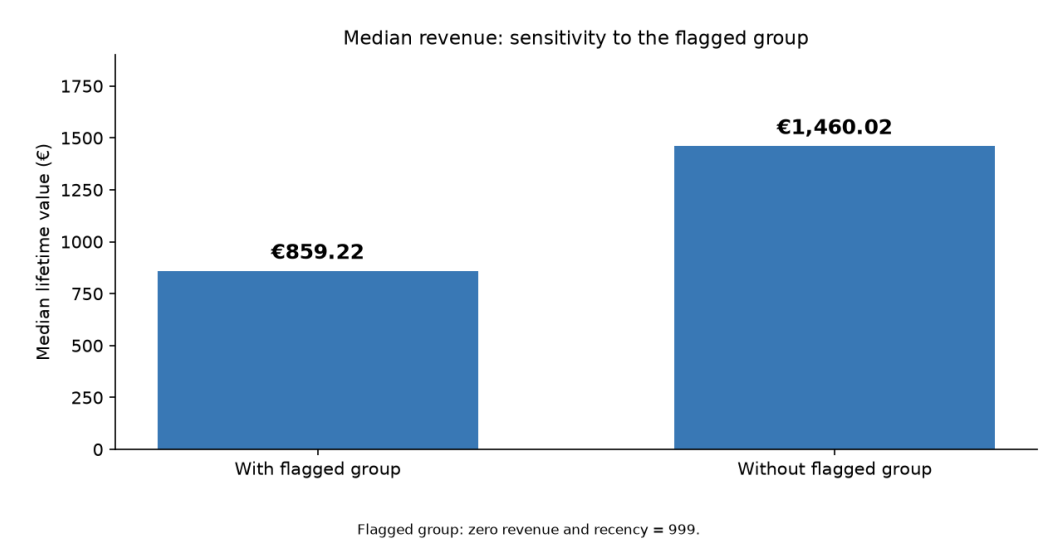
unique members
15 exact duplicates removed

1,040 missing + 82 negative

Revenue values excluded only from economic comparisons.

1,296 members flagged

Zero revenue and recency = 999. All retained.



This pattern does not prove inactivity or justify removing members.

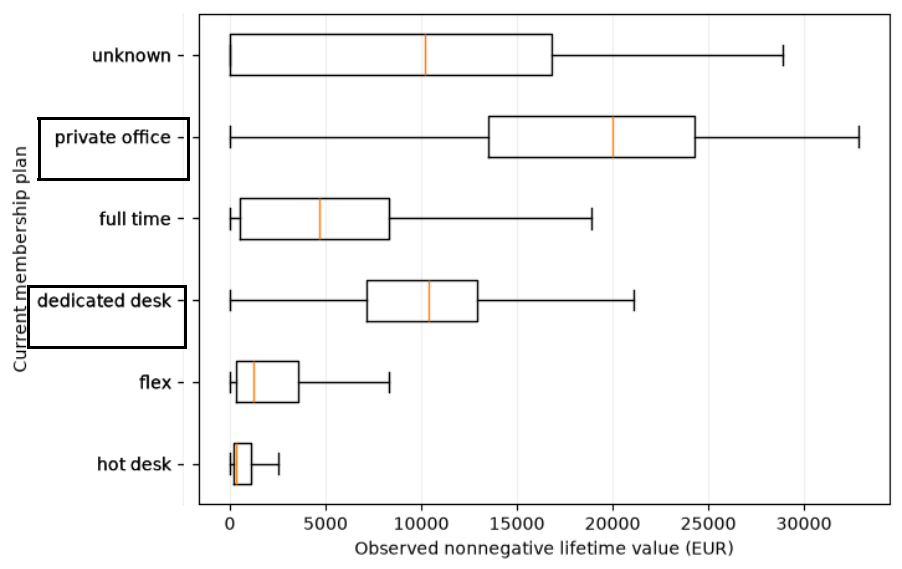
Economic view: n = 14,328 with the flagged group and 13,032 without it. Original member records remain in the dataset.

Higher value does not imply higher utilization

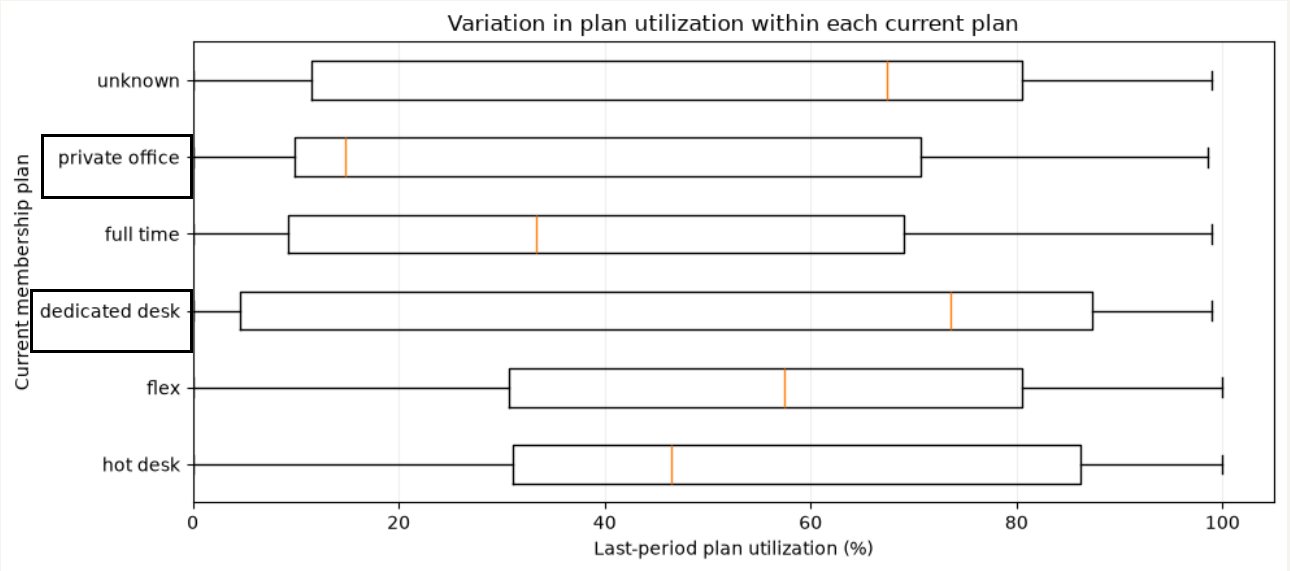
Private office €20,000 median revenue / 14.79% median utilization

Dedicated desk €10,379 median revenue / 73.60% median utilization

Lifetime revenue (€)



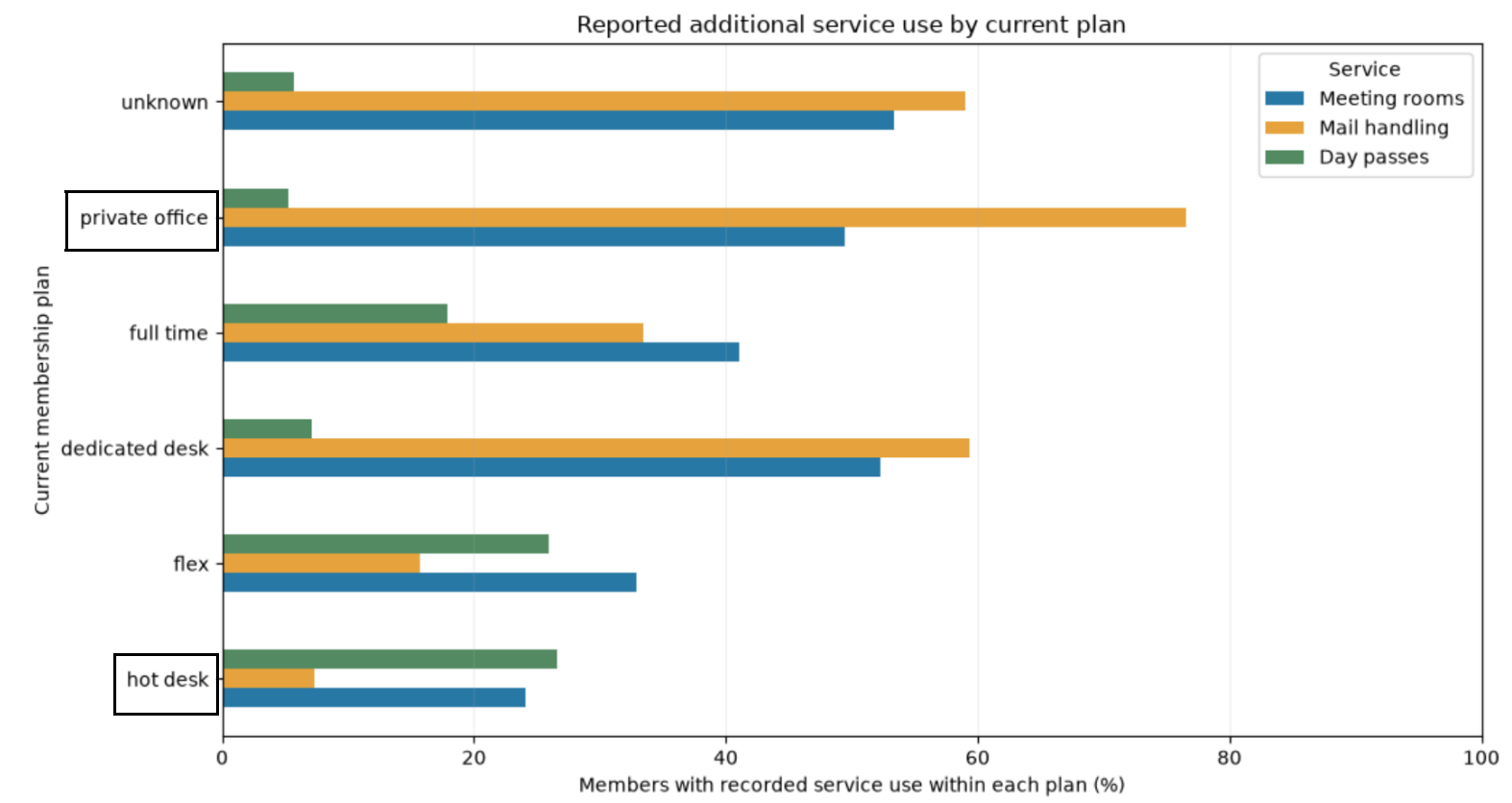
Last-period utilization (%)



Value and usage warrant separate segmentation perspectives.

Revenue and utilization use different time windows and available observations.
Boxes show the middle 50% and median. Revenue outlier markers are hidden, not removed.

Services reveal different member needs



Private office VS hot desk

76.5% vs 7.4%

Mail handling

5.3% vs 26.6%

Day passes

≈20%

Have no recorded use of the nine services

Business support and flexible access are useful clues to member needs.

Percentages use all members within each current plan. Services can overlap. Flags describe past use, not intensity.
Existing meeting-room flag conflicts remain. These are segmentation clues, not confirmed segments.

New features add context, but may overlap

utilization_rate

Actual visits / entitled visits

Share of the last-period allowance used.
10 undefined cases with zero allowance.

n_services_used

Number of service flags marked True

Variety across nine services.
Does not measure usage intensity.

Potential redundancy: Spearman correlation

Weekly visits and last-period visits	**0.933**
Utilization and weekly visits	**0.840**

Correlated features may give attendance too much weight in clustering.

Compare feature sets during modeling. Service history and last-period utilization cover different time windows.
Spearman uses available pairs: n = 15,450 and 15,440, respectively.

Three broad profiles to investigate

Initial hypothesis: three broad profiles to validate.
The final number of clusters remains open.

01

Basic or flexible access

Lower attendance
Basic or flexible access needs

02

Frequent space users

Higher attendance
More regular use of the space

03

High value, lower presence

Higher economic value
Business-support services

Main challenges Unclear dates, extreme values, missing data and redundant measures.

Next: compare methods, cluster counts and stability across perspectives.